Antisymmetrized Geminal Power Coherent States

Brian Weiner

Department of Physics, Pennsylvania State University, DuBois PA 15801

A detailed analysis of electronic coherent states associated with the one particle unitary group is presented in terms of coset generators of this group. For even number of electrons these states form non linear manifolds of Antisymmetrized Geminal Powers, that are parameterized by complex coordinates. Different classes of manifolds of Antisymmetrized Geminal Power states are associated with distinct coset decompositions and produce different restrictions of the time dependent Schrödinger equation.

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I. INTRODUCTION

Isolated molecular systems do not interact with their environment and are thus unobservable. In order to have a physical effect and be observed they must respond to external probes, such as electromagnetic fields, and form a composite interacting system whose evolution is described by a joint Hamiltonian. An example of this is the effect of radiation on a molecular subsystem, which can be monitored by changes in the molecules density and reduced density matrices. If the radiation is weak the behavior of the interaction is well described by the linear response of the molecular system to the radiation, i.e. is linearly proportional to the radiation strength, while if the radiation is intense the molecular system responds in a nonlinear way. The interaction of the electromagnetic field with molecules can be approximated in a standard semi classical fashion by averaging over the photonic degrees of freedom to produce effective one electron observables. This leads to a reduced Hamiltonian for the molecular subsystem that includes perturbing effective time dependent one body potentials which describe the interaction with the radiation field. In the case of weak intensity radiation, where the linear response regime is sufficient to describe the changes in the molecular system, one needs to only consider responses that are first order in these observables.

The temporal behavior of a state of physical system is completely described by the evolution of its full density matrix, which in term determines the evolution of all of its reduced density matrices. These variations are conveniently expressed in terms of various propagators, which thus provide a comprehensive formulation of the response of the molecular subsystem to external probes. As is well known exact solutions for quantum evolutions have only been determined for simple model systems, and are not known for multi-electron molecular systems that involve coulombic interactions between electrons. Thus for practical applications the goal is to develop systematic, cost effective, theoretically well founded and non arbitrary approximation schemes, that are accurate enough to explain experimental results. A major focus of the illustrious career of professor Osvaldo Goscinski has been the analysis and construction of approximate propagators, that fulfill these objectives. In particular the one-electron and polarization propagators, where he effectively used the properties of super operators acting on operator manifolds associated with a given reference state to generate various orders of approximation [1][2][3][4][5][6][7]. In this article I will concentrate, for illustrative purposes, on the polarization propagator for a molecular system with a fixed even number of electrons at a fixed geometry. This propagator describes the linear response of the First Order Reduced Density Matrix (FORDM) to electromagnetic radiation.

The polarization propagator is a hermitian matrix of time dependent response functions of the matrix elements of the FORDM. If one takes the

Laplace transform of this propagator and expresses the resulting matrix in diagonal form, in terms of normal mode operators, one can clearly observe that the poles of this propagator are excitation energies and the associated residues the transition probabilities for excitations generated by one particle operators between stationary states, $\{|\Psi_k\rangle\}$, of the molecular system. If the initial molecular reference state is stationary (usually the ground state, but not necessarily so) then the poles are the excitation energies and the residues the transition probabilities from that state. If the transition probability is 100% for a given one particle operator then this operator produces an excited state from the initial stationary state and is called a state excitation operator. If such an operator produces the excited state $|\Psi_k\rangle$, then its effect on the reference state $|\Psi_0\rangle$ is the same as the N-particle bra-ket operator $|\Psi_k\rangle\langle\Psi_0|$.

Approximations to the polarization propagator are often based on following criteria,

- all one particle operators that diagonalize the propagator matrix are state excitation operators,
- their adjoints annihilate the reference state,
- the reference state is stationary with respect to at least one particle variations.

If an approximation satisfies these conditions one says that it is a consistent approximation, however, in general, usually one or more of these conditions are not satisfied and the approximation is inconsistent. Some approximations can even lead to propagators that do not correspond to a valid N-particle reference state and thus are not even N-representable.

An important point of note is that it is not necessary that the normal mode one-particle operators actually produce excited states from the reference state, in fact, in the exact case they generally do not. Hence there is thus some level of ambiguity about polarization propagator approximations, if one is trying to obtain an approximation to a one-particle linear response function then the annihilation condition is not necessary, though stationarity and N-representability are desirable, while if one is asserting that the polarization propagator is a one-particle approximation to the full N-particle propagator and is used to produce excited states, as well as excitation energies and transition probabilities, then the annihilation condition is necessary for consistency.

In the late 1970's it was found [8][5][6][9][10] that an Antisymmetrized Geminal Power (AGP) [11] state was the most general $2M$ electron state that was annihilated by a maximal set of one electron operators, (for odd number of electrons Generalized AGP (GAGP) states [12] are needed but their properties can be derived from AGP states). The $2M$-electron AGP $|g^M\rangle$ state is an antisymmetrized product of M two particle geminals

$$|g\rangle = \sum_{1\leq j<k\leq s} g_{jk}\,|\varphi_j\varphi_k\rangle \tag{1}$$

where $\{g_{ij}\}$ are complex coefficients and $\{\,|\varphi_j\rangle|\,1 \leq j \leq r = 2s\}$ is an orthonormal basis for the one particle Hilbert space $\mathcal{H}^1$. However even when the reference state was an energy optimized AGP state the polarization propagator approximation failed to be consistent, in that the adjoints of the normal mode excitation operators did not annihilate the reference state, although the numerical results obtained from the application of this approximation scheme agreed well with experimental values of excitation energies and dipole induced transition probabilities, i.e. oscillator strengths, for example [13][14][15][16][17][18][19][20][21][22][23][24][25].

It turns out that the annihilation condition can be naturally expressed in terms of group Coherent States (CS) [26][27][28], where the reference state corresponds to a lowest or highest weight state associated with a representation of the group. The basic ingredients needed for the construction of a family of group CS's in a vector space is the existence of an irreducible representation of a group and a one dimensional representation of a subgroup. The subgroup is called the isotropy or invariance group of the CS's and the family of CS's is generated by the action of the coset, formed by factoring the group with respect to (wrt) this subgroup, on the reference state. In the study of electronic systems it is natural to consider the real, compact, [29] Lie group $U(r)$ of unitary transformations of the one electron state space $\mathcal{H}^1$ (of dimension $r = 2s$), that has irreducible representations on all N-electron state spaces $\mathcal{H}^N$. The family of CS's in this case are nonlinear manifolds, $\mathcal{M}$, of the N-electron Hilbert space, $\mathcal{H}^N$, generated by cosets formed by one-particle operators.

One can construct approximate propagators based on stationary states of the restriction of the Time Dependent Schrödinger Equation (TDSE) to these manifolds. Solutions to such restrictions of the TDSE are obtained by using the Time Dependent Variational Principle (TDVP) [30] or equivalently the Stationary Phase Approximation (SPA) [31] to the path integral expressed in terms of paths on this CS manifold. The resulting Equations of Motion (EOM) are those of a generalized *classical* system described by a curved phase space, a consequence of the fact that a family of group coherent states forms a differentiable manifold [32] whose tangent spaces have an intrinsic symplectic structure [29]. Approximate stationary states of the molecular system then correspond to the special stationary phase "paths" that are critical points of the corresponding classical Hamiltonian. The linear response of these approximate stationary states to external perturbing fields then involves the generators of the Lie algebra of the group, and hence describes the response of the reduced density matrix formed by these operators. Thus the response of the FORDM is determined by generators of one-particle groups.

In this article we will describe how different coset decompositions of the one particle unitary group $U(2s)$ correspond to different CS manifolds of AGP states and how these manifolds can be parameterized by the coordinates of the subspaces of the direct sum Lie algebra expansions adapted to these coset decompositions. These CS manifolds form generalized curved phase spaces in which classical EOM approximating the TDSE can be formulated, leading to a classical linear response theory of time dependent perturbations that extends and clarifies certain aspects of the quantum Random Phase Approximation and propagator approximations. However due to space limitations these considerations will be developed in future manuscripts.

In section II we discuss the Lie algebra structure of Fermi-Fock space, which is the direct sum of all of the state spaces $\{\mathcal{H}^N\}$, in section III we will describe the possible decompositions of the Lie algebra $u\,(2s)$ of $U(2s)$, the one particle group associated with $\mathcal{H}^1$, and thus the factorizations of $U(2s)$ that lead to manifolds of coherent states in $\mathcal{H}^N$ that can be based on $U(2s)$. In section IV we study in detail the forms of AGP coherent states. We also show how the coherent state construction applied to the unitary group of N-electron space can reproduce the familiar configurational interaction expansion. We conclude with a summary and discussion in section V.

II. LIE ALGEBRAS

The most convenient representation of the generators of the Lie algebra of $U(2s)$ for fermionic quantum mechanics is by second quantized operators acting in Fermi-Fock space, $\mathcal{H}_F$, which is defined to be the direct sum

$$\mathcal{H}_F = \bigoplus_{0\leq N\leq r} \mathcal{H}^N, \tag{2}$$

where

$$\mathcal{H}^N = \bigwedge \mathcal{H}^1 \tag{3}$$

is the N -fold antisymmetric tensor product of the one particle state space $\mathcal{H}^1$, a space generated by an orthonormal basis of one particle functions of position and spin

$$\{\varphi_j |\, 1 \leq j \leq r\}\,. \tag{4}$$

The Banach space [29] of bounded operators acting in $\mathcal{H}_F$ is denoted by $\mathcal{B}\left(\mathcal{H}_F\right)$, and is spanned by a second quantized operator basis

$$\left\{ a^\dagger_{j_1} \cdots a^\dagger_{j_p} a_{k_q} \cdots a_{k_1} \middle|\, 1 \leq j_1 < \cdots j_p \leq r; 1 \leq k_1 < \cdots k_p \leq r; 1 \leq p, q \leq r \right\}, \tag{5}$$

where the second quantized operators $\left\{a_j^\dagger, a_j\right\}$ act on the zero particle vacuum $|\phi\rangle$ in the following fashion

$$\begin{aligned} a_j\,|\phi\rangle &= 0 \\ a_j^\dagger\,|\phi\rangle &= |\varphi_j\rangle \end{aligned} \tag{6}$$

and satisfy the Canonical Anti-commutation Relationships (CAR)

$$\left[a_j^\dagger, a_k\right]_+ = \delta_{jk} I. \tag{7}$$

The operator space $\mathcal{B}(\mathcal{H}_F)$ contains realizations of many classical Lie groups, Lie algebras and universal enveloping algebras [29]. In particular the generators

$$\left\{a_j^\dagger, a_k, a_j^\dagger a_k - \frac{1}{2}\delta_{jk} I, a_j^\dagger a_k^\dagger, a_j a_k; 1 \le j \le k \le r\right\} \tag{8}$$

satisfy the commutation relationships of the Lie algebra $so\,(2r+1, \mathbb{R})$, (see [28] for example for the definitions of classical Lie groups and algebras), and generate a representation of the universal enveloping algebra $so\,(2r+1, \mathbb{R})$ in $\mathcal{B}(\mathcal{H}_F)$, which is fact all of $\mathcal{B}(\mathcal{H}_F)$. The subset of these generators

$$\left\{a_j^\dagger a_k - \frac{1}{2}\delta_{jk} I, a_j^\dagger a_k^\dagger, a_j a_k; 1 \le j \le k \le r\right\} \tag{9}$$

satisfy the commutation relationships of the Lie algebra of $so\,(2r, \mathbb{R})$ and generate a representation of the universal enveloping algebra of $so\,(2r, \mathbb{R})$ in $\mathcal{B}(\mathcal{H}_F)$. The subset of these generators

$$\left\{a_j^\dagger a_k - \frac{1}{2}\delta_{jk} I; 1 \le j \le k \le r\right\} \tag{10}$$

satisfy the commutation relationships of the Lie algebra $u\,(r)$ and generate a representation of the universal enveloping algebra of $u\,(r)$ in $\mathcal{B}(\mathcal{H}_F)$.

The representation of $so\,(2r+1, \mathbb{R})$ is irreducible, while the representations of $so\,(2r, \mathbb{R})$ and $u\,(r)$ are reducible. The irreducible representations of $so\,(2r, \mathbb{R})$ are obtained by restricting to the subspace of even or odd number states in $\mathcal{H}_F$, while the irreducible representations of $u\,(r)$ are obtained by restricting to subspaces of states of fixed integer particle number.

III. ONE PARTICLE GROUPS

In this section we describe the possible direct sum decompositions of the Lie algebra, $u(2s)$, of the group $U(2s)$ of unitary transformations of one particle state space, $\mathcal{H}^1$ of dimension $2s$, which lead in a canonical fashion to families of CS's in N-particle state space $\mathcal{H}^N$. These decompositions are based on the subgroup of special orthogonal transformations, $SO(2s, \mathbb{R})$, of a real $2s$-dimensional Hilbert space, the subgroups, $U(N) \times U(2s-N)$, formed by products of unitary transformations of complex N and $2s-N$ dimensional Hilbert spaces and the subgroup, $USp(2s)$, of transformations of a complex $2s$-dimensional Hilbert space, that are simultaneously unitary and symplectic. This latter subgroup is isomorphic to the group of unitary transformations, $U(s, \mathbb{Q})$ of a s-dimensional quaternionic Hilbert space. We signify the CS's that they induce as G/K-CS's, where for example $G = U(2s)$ and the subgroup $K \in \{SO(2s, \mathbb{R}), U(N) \times U(2s-N), USp(2s)\}$.

The Lie algebra $u(2s)$ has three direct sum Cartan decompositions [29] which are in terms of the Lie algebras, $so(2s, \mathbb{R})$, $[u(N) \oplus u(2s-N)]$, and $usp(2s)$, of its maximal compact [29] subgroups $SO(2s, \mathbb{R})$, $[U(N) \times U(2s-N)]$, and $USp(2s)$ respectively, (see [28] for the definitions of these groups). These decompositions are

$$u(2s) = so(2s, \mathbb{R}) \oplus so(2s, \mathbb{R})^{\perp} \tag{11a}$$

$$u(2s) = [u(N) \oplus u(2s-N)] \oplus [u(N) \oplus u(2s-N)]^{\perp}; \quad 0 \leq N \leq 2s \tag{11b}$$

$$u(2s) = usp(2s) \oplus usp(2s)^{\perp} \tag{11c}$$

where the orthogonality is with respect to the Killing form inner product [29] of $u(2s)$. The decomposition eqs. $(11a, 11b, 11c)$ define the following distinct coset decompositions of $U(2s)$

$$U(2s)/U(N) \times U(2s-N)\ ; \quad 0 \leq N \leq 2s \tag{12a}$$

$$U(2s)/USp(2s) \tag{12b}$$

$$U(2s)/SO(2s, \mathbb{R}) \tag{12c}$$

and thus three classes of families of coherent states (see Section IV).

The motivation of using *maximal* compact subgroups in eqs. $(12a), (12b)$ and $(12c)$ is to produce coset spaces that are irreducible Riemannian Manifolds [32] (a Riemannian Manifold has tangent spaces that are inner product spaces), i.e. they cannot be factored into a Riemannian product of smaller Manifolds. This technical feature, although important in the classification of Riemannian Manifolds, is too restrictive when considering manifolds of CS's, where we also need to examine decompositions generated by subgroups of these maximal compact groups. We thus enlarge the designation G/K-CS to include subgroups of K.

In order for any of the subgroups

$$\{SO(2s, \mathbb{R}), \{U(N) \times U(2s-N) |\, 0 \leq N \leq 2s\}, USp(2s)\}, \tag{13}$$

or their subgroups, to be isotropy groups of a given N-electron state, the state must carry a one dimensional unitary representation of that subgroup and not of any larger subgroup containing that subgroup.

There are *no* states in $\mathcal{H}^N$ that produce a one-dimensional representation of $SO(2s,\mathbb{R})$ and we conjecture that there are *no* subgroups of $SO(2s,\mathbb{R})$, which *are not* subgroups of $U(N)\times U(2s-N)$ or $USp(2s)$, that produce one-dimensional representations on $\mathcal{H}^N$. Thus there are *no* CS manifolds in $\mathcal{H}^N$ based on subgroups of $SO(2s,\mathbb{R})$ that cannot be described in terms of $U(N)\times U(2s-N)$ or $USp(2s)$ CS's.

The only subgroup from the set $\{U(N)\times U(2s-N)|\, 0\leq N\leq 2s\}$ that produces a one-dimensional representation in $\mathcal{H}^N$ is $U(N)\times U(2s-N)$. These $U(2s)/U(N)\times U(2s-N)$ -coherent states give the well known Thouless parameterization [33] of the manifold of Independent Particle States (IPS) and lead to the standard Random Phase Approximation (RPA) for the linear response function. The subgroup $U(N)\times U(2s-N)$ is actually a subgroup of $USp(2s)$ and we display in section (IV) that $U(2s)/[U(N)\times U(2s-N)]$-CS's can in fact be described in terms of the class of $U(2s)/USp(2s)$-CS's.

We show by construction in section (IV) that the group $USp(2s)$ and its subgroups do have one dimensional representations in $\mathcal{H}^N=\mathcal{H}^{2M}$ and one can form

$$U(2s)/K - \text{coherent states, where } USp(2s)\supseteq K, \tag{14}$$

which are manifolds of general Antisymmetrized Geminal Power (AGP) states (see Appendix A). All manifolds of CS's in $\mathcal{H}^{2M}$ based on $U(2s)$ can be expressed in terms of the cosets appearing in eq. (14), which have the form

$$U(2s)/[USp(2\omega_1)\cdots\times USp(2\omega_p)\times SU(2)^{\sigma}\times SU(2\xi)]\ , \tag{15}$$

where

$$SU(2)^{\sigma} = \overbrace{SU(2)\times\cdots\times SU(2)}^{\sigma-\text{factors}}, \tag{16}$$

the index set $\{\sigma,\omega_1,\cdots,\omega_\rho,\xi\}$ characterizes the type of AGP reference state and its elements satisfy

$$\begin{aligned} 2\sigma+2\omega+2\xi &= r = 2s \\ \omega &= \sum_{1\leq j\leq\rho}\omega_j. \end{aligned} \tag{17}$$

The index set $\{\sigma,\omega_1,\cdots,\omega_\rho,\xi\}$ is determined by the array $\{c_j|\, 1\leq j\leq s\}$ of canonical coefficients [**?**] of the geminal that determines the AGP state according to:

- σ = number of non zero, non equal $\{c_j\}$'s

- ω_j = number of equal coefficients in the j^{th} set
- ρ = number of sets of equal $\{c_j\}$'s
- ξ = number of zero coefficients.

We now consider a few particular examples. If all the coefficients are different and non zero, so that $\xi = 0, \omega = 0, \rho = 0$, the isotropy subgroup is

$$SU(2)^s = \overbrace{SU(2) \times \cdots \times SU(2)}^{s-\text{factors}}, \tag{18}$$

which is the smallest one possible, leading to the biggest coset. In this case the action of coset on the reference AGP generates the manifold containing all AGP states.

If $\xi \neq 0$ the reference AGP is produced by a degenerate geminal and the action of the associated coset only produces a submanifold of AGP states, all of which correspond to geminals that have the same null space (see Appendix A) as the reference AGP.

If $\rho \neq 0$ then the geminal has some extreme components and again the coset only generates a sub manifold of AGP's from the reference.

If $\sigma = 0$, $\xi = 0$ and $\rho = s$ then the reference state is an extreme AGP (see Appendix A) and the action of the coset on the reference produces the manifold of *all* extreme AGP states.

All CS manifolds in $\mathcal{H}^{2M}$ based on the one particle group $U(2s)$ are families of AGP states, some of these manifolds are irreducible Riemannian manifolds and correspond to cosets formed by the maximal compact subgroups $U(2M) \times U(2s-2M)$ and $USp(2s)$, while others are reducible and correspond to non maximal compact subgroups $USp(2\omega_1) \times \cdots \times USp(2\omega_p) \times SU(2)^{\sigma} \times SU(2\xi)$. Each of these manifolds of CS's have certain basic physical properties e.g. $U(2M) \times U(2s-2M)$ invariant manifold describes uncorrelated Independent Particle States (IPS's), the $USp(2s)$ invariant manifold describes highly correlated extreme AGP states that are superconducting, while the $USp(2\omega_1) \times \cdots \times USp(2\omega_\rho) \times SU(2)^{\sigma} \times SU(2\xi)$ invariant manifold for general $\{\omega_1, \cdots, \omega_\rho, \sigma, \xi\}$ describe intermediate types of correlation and linear response properties, see for a particular example [34], most of which have not been explored in any depth.

IV. COHERENT STATES

A. Definition and Construction

The coherent state construction is based on the Hilbert space of states $\mathcal{H}$ that carries unitary irreducible representations $\mathcal{R}(G)$ of a compact group

G on $\mathcal{H}$ and a one dimensional representation of a subgroup $K \subset G$. The groups G we consider are $U\left(\mathcal{H}^N\right)$ and $U\left(\mathcal{H}^1\right)$ the groups associated with N-electron state space and 1-electron state space respectively. In the finite dimensional case these groups are the classical matrix groups and $U\left(\mathcal{H}^N\right) \equiv U\left(\binom{r}{N}\right)$ and $U\left(\mathcal{H}^1\right) \equiv U\left(r\right)$, in general we will use the matric notation even though in an exact treatment $r = \infty$ and one should be aware of possible infinite dimensional pathologies. The irreducible representations we consider are $\mathcal{R}\left(U\left(\binom{r}{N}\right)\right)$ and $\mathcal{R}\left(U\left(r\right)\right)$ that are carried by $\mathcal{H}^N$, unless otherwise stated we will not distinguish between the groups G and K and their representations on $\mathcal{H}^N$.

A state $|\Phi\rangle \in \mathcal{H}^N$ is a reference state for a group G if there is an isotropy subgroup K, that is contained in G, that has 1-dimensional representation on $\mathcal{H}^N$ defined by

$$K = \left\{ g| \quad \mathcal{R}\left(g\right)|\Phi\rangle = e^{i\theta(g)}\,|\Phi\rangle\,; g \in G; \theta\left(g\right) \in \mathbb{R}\right\}. \tag{19}$$

If such isotropy groups exists G can be factored as

$$G = \left(G/K\right)K. \tag{20}$$

B. AGP Coherent States

The group $U\left(2s\right)$ has subgroups that leave reference $2M$-electron AGP states, $\left|g^N\right\rangle$ invariant up to an overall phase factor and thus satisfy eq. (19). AGP states are given [**?**] by

$$\left|g^N\right\rangle = \sum_{1\leq j_1<j_2<\cdots<j_N\leq s} c_{j_1}\cdots c_{j_N}\,|\eta_{j_1}\eta_{j_1+s}\cdots\eta_{j_N}\eta_{j_N+s}\rangle\,, \tag{21}$$

where $\{\,|\eta_j\rangle|\, 1 \leq j \leq r\}$ is a complete orthonormal basis for $\mathcal{H}^1$, $\{\,c_j|\, 1 \leq j \leq s\}$ are positive real numbers called the canonical coefficients of a two electron geminal $|g\rangle$ that has the canonical expansion

$$|g\rangle = \sum_{1\leq j\leq s} c_j\,|\eta_j\eta_{j+s}\rangle\,. \tag{22}$$

The state $\left|g^N\right\rangle$ can also be expressed in terms of the coset $SO\left(2r,\mathbb{R}\right)/SU\left(r\right)$ of the group $SO\left(2r,\mathbb{R}\right)$ acting on the zero particle vacuum state $|\phi\rangle$ of $\mathcal{H}_F$ [35] as

$$\left|g^N\right\rangle = P_N \exp\left\{\sum_{1\leq j\leq s} c_j b_j^\dagger b_{j+s}^\dagger\right\}|\phi\rangle\,, \tag{23}$$

where P_N is the orthogonal projector onto $\mathcal{H}^N$ and $\left\{ b_j^\dagger \middle| 1 \leq j \leq s \right\}$ are the creators for the one particle states $\{ |\eta_j\rangle | 1 \leq j \leq r\}$ i.e.

$$b_j^\dagger |\phi\rangle = |\eta_j\rangle . \tag{24}$$

This form has been explored in some detail in [36]. However we will not pursue this form of the AGP state further in this article.

C. Coset Factorization for $U(2s)$

In general the action of $U(2s)$ on a reference AGP, $|g^N\rangle$, state produces a manifold, $\mathcal{M}(\sigma, \omega_1, \cdots, \omega_\rho, \xi)$ of AGP states given by

$$\mathcal{M}(\sigma, \omega_1, \cdots, \omega_\rho, \xi) = \{[U(2s)/K] \times K\} |g^N\rangle , \tag{25}$$

where

$$K = USp(2\omega_1) \cdots \times USp(2\omega_p) \times SU(2)^\sigma \times SU(2\xi) \tag{26}$$

and the geminal is of type $(\sigma, \omega_1, \cdots, \omega_\rho, \xi)$. Each element of the manifold can be expressed as

$$\left| g(c)^N \right\rangle = c k_{\omega_1} \cdots k_{\omega_\rho} k_\sigma k_\xi \left| g^N \right\rangle = c \left| g^N \right\rangle , \tag{27}$$

where the isotropy subgroup is given by the products $k_{\omega_1} \cdots k_{\omega_\rho} k_\sigma k_\xi$, where

$$k_{\omega_j} \in USp(2\omega_j) ; 1 \leq j \leq \rho, k_\sigma \in SU(2)^\sigma , k_\xi \in SU(2\xi) , c \in U(2s)/K. \tag{28}$$

The most general AGP manifold is produced when $\sigma = s$, ($\Leftrightarrow (\rho = 0 \ \& \ \xi = 0)$), so it is advantages to first consider that case.

D. $SU(2)^s$ factors

The isotropy subgroup, K, for an AGP reference state that has non equal and nonzero canonical coefficients $\{ c_j | 1 \leq j \leq s\}$, i.e. $\sigma = s$ and $|g\rangle$ belongs the class $\{s, 0, 0\}$, is the product group $SU(2)^s$, which produces the coset decomposition

$$U(2s) = [U(2s)/SU(2)^s] \times SU(2)^s . \tag{29}$$

One can express coset representatives in a factored form as the product $G_{+1}G_0G_{-1}$ [37], where $\{G_{+1}, G_0, G_{-1}\}$ are subgroups of the linear group $GL(2s, \mathbb{C})$, which is the complexification of $U(2s)$, and the group G_{-1} leaves the reference AGP state invariant (note that it is different to the real compact isotropy group). Restricting attention to the connected components of these subgroups, that contain the identity element, one can express this factorization in terms of Lie algebras as

$$U\left(2s\right)=e^{N_+}e^{N_0}e^{N_-}e^{\mathcal{K}}, \tag{30}$$

where

$$\mathcal{K}=\overbrace{su\left(2\right)\oplus\cdots\oplus su\left(2\right)}^{s\text{-subspaces}}, \tag{31}$$

$\{N_{+1},N_0,N_{-1}\}$ are Lie subalgebras of $gl\left(2s,\mathbb{C}\right)$, the Lie algebra of $GL\left(2s,\mathbb{C}\right)$. The subalgebras $\{N_{-1},\mathcal{K}\}$ annihilate $\left|g^N\right\rangle$ i.e.

$$b\left|g^N\right\rangle=0\quad\forall b\in N_{-1}\oplus\mathcal{K}, \tag{32}$$

while the subalgebras $\{N_{+1},N_0\}$ have the property

$$b\left|g^N\right\rangle\neq0\quad\forall b\in N_{+1}\oplus N_0. \tag{33}$$

Introducing coordinate systems $\left\{\mathbf{z}_+,\lambda,\mathbf{z}_-,\eta\right\}$ with respect to fixed bases of $\{N_{+1},N_0,N_{-1},\mathcal{K}\}$, that depend on the reference AGP state, one can parameterize group elements $u\in U\left(2s\right)$ as

$$u\left(\tilde{\mathbf{z}},\eta\right)=\tilde{c}\left(\tilde{\mathbf{z}}\right)k\left(\eta\right)=c_{+1}\left(\mathbf{z}_+\right)c_0\left(\lambda\right)c_{-1}\left(\mathbf{z}_-\right)k\left(\eta\right), \tag{34}$$

where $\tilde{\mathbf{z}}\equiv\left(\mathbf{z}_+,\lambda,\mathbf{z}_-\right)$. The action of $U\left(2s\right)$ on the AGP state is

$$\begin{aligned}u\left(\tilde{\mathbf{z}},\eta\right)\left|g^N\right\rangle&=\tilde{c}\left(\tilde{\mathbf{z}}\right)k\left(\eta\right)\left|g^N\right\rangle\\&=c_{+1}\left(\mathbf{z}_+\right)c_0\left(\lambda\right)c_{-1}\left(\mathbf{z}_-\right)k\left(\eta\right)\left|g^N\right\rangle=c_{+1}\left(\mathbf{z}_+\right)c_0\left(\lambda\right)\left|g^N\right\rangle,\end{aligned} \tag{35}$$

showing that the non redundant coordinates $\mathbf{z}\equiv\left(\mathbf{z}_+,\lambda\right)$ describe a configuration space, that is a sub manifold $\mathcal{M}$ (in general nonlinear) of $\mathcal{H}^N$ with a coordinate system $\{\mathbf{z}\}$, that parameterize the set of $\{\left|\Phi\left(\mathbf{z}\right)\right\rangle\}$ of $U\left(2s\right)/SU\left(2\right)^s$-coherent states

$$\left|\Phi\left(\mathbf{z}\right)\right\rangle=c_{+1}\left(\mathbf{z}_+\right)c_0\left(\lambda\right)\left|g^N\right\rangle\equiv c\left(\mathbf{z}\right)\left|g^N\right\rangle \tag{36}$$

and at the same time represents the coset space $U\left(2s\right)/SU\left(2\right)^s$.

The AGP state dependent bases one chooses are

$$N_{+1} = \mathbb{C}-\text{ linspan}\Big\{c_j a_k^\dagger a_j - \sigma(j)\,\sigma(k)\,c_k a_{-j}^\dagger a_{-k}; \qquad -s \le j < k \le s, |j| \neq |k|\,;\ \sigma(j) = \text{sign}(j)\} \tag{37a}$$

$$N_0 = \mathbb{C}-\text{ linspan}\left\{\left(a_j^\dagger a_j + a_{-j}^\dagger a_{-j}\right), \left(a_j^\dagger a_{-j} + a_{-j}^\dagger a_j\right), \left(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}\right), \quad i\left(a_j^\dagger a_{-j} - a_{-j}^\dagger a_j\right); 1 \le j \le s\right\} \tag{37b}$$

$$N_{-1} = \mathbb{C}-\text{ linspan}\Big\{c_k a_k^\dagger a_j + \sigma(j)\,\sigma(k)\,c_j a_{-j}^\dagger a_{-k} \qquad -s \le j < k \le s, |j| \neq |k|\,;\ \sigma(j) = \text{sign}(j)\} \tag{37c}$$

$$\mathcal{K} = \mathbb{R}-\text{ linspan}\left\{i\left(a_j^\dagger a_{-j} + a_{-j}^\dagger a_j\right), i\left(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}\right), \quad \left(a_j^\dagger a_{-j} - a_{-j}^\dagger a_j\right)\right\}; \quad 1 \le j \le s, \tag{37d}$$

where we have translated the indexing of the Canonical General Spin Orbitals (CGSO's) by

$$|\eta_j\rangle \to |\eta_{j-s}\rangle \quad 1 \le j \le 2s. \tag{38}$$

It is useful to note that

$$N_0 = \{\mathcal{K} \oplus \mathcal{N}\}^{\mathbb{C}}, \tag{39}$$

where

$$\mathcal{N} = \bigoplus_{1 \le j \le s}\left\{\mathbb{R}-\text{ linspan}\left\{\left(a_j^\dagger a_j + a_{-j}^\dagger a_{-j}\right)\Big| \quad 1 \le j \le s\right\}\right\} \tag{40}$$

and $\{\}^{\mathbb{C}}$ denotes complexification. The factors in the group decomposition are

$$c_{+1}(\mathbf{z}_+) = \exp\left\{\sum_{\substack{-s \le j < k \le s \\ |j| \neq |k|}} z_{+jk}\left\{c_j a_k^\dagger a_j - \sigma(j)\,\sigma(k)\,c_k a_{-j}^\dagger a_{-k}\right\}\right\} \equiv \exp \sum_{\substack{-s \le j < k \le s \\ |j| \neq |k|}} z_{+jk} q_{jk}^\dagger, \tag{41}$$

$$c_{-1}(\mathbf{z}_-) = \exp\left\{\sum_{\substack{-s \le j < k \le s \\ |j| \neq |k|}} z_{-jk}\left\{c_k a_k^\dagger a_j + \sigma(j)\,\sigma(k)\,c_j a_{-j}^\dagger a_{-k}\right\}\right\} \equiv \exp \sum_{\substack{-s \le j < k \le s \\ |j| \neq |k|}} z_{-jk} q_{jk}, \tag{42}$$

$$c_0(\lambda) \equiv \exp\left\{ \sum_{\substack{1\leq j<\leq s\\ 1\leq k\leq 4}} \lambda_{jk} n_{jk} \right\}, \tag{43}$$

and

$$h(\eta) \equiv \exp\left\{ \sum_{\substack{1\leq j<\leq s\\ 1\leq k\leq 3}} \eta_{jk} u_{jk} \right\}, \tag{44}$$

where $\{z_{+jk}, z_{-jk}, \lambda_{jk}\} \in \mathbb{C}$, $\{\eta_{jk}\} \in \mathbb{R}$, and $\{n_{jk}, u_{jk}\}$ are given by eqs. $(37b)$ and $(37d)$ respectively.

In order that the transformation $c_{+1}(\mathbf{z}_+)\, c_0(\lambda)\, c_{-1}(\mathbf{z}_-)\, k(\eta)$ be unitary the values of $\{\mathbf{z}_+, \lambda, \mathbf{z}_-\}$ must be constrained resulting in dependencies between the variables. The action of such unitary transformations on a normalized reference state produce normalized coherent states that can be expressed in terms of the coset representatives $c_{+1}(\mathbf{z}_+)\, c_0(\lambda)$. Some of the dependencies between the variables $\{\mathbf{z}_+\}$ and $\{\lambda\}$ can be avoided if one considers unnormalized coherent states by modifying the coset representative $c_0(\lambda)$ and explicitly considering the norm of the state in various expectation value expressions.

E. $USp(\omega_j)$ factors

The changes from the $SU(2)^s$ case can be illustrated by an $\sigma = s-2, \rho = 1, \omega_1 = 2$ example (in this case $|g\rangle$ belongs to the class $\{s-2, 2, 0\}$), which clearly leads to the general picture for $\rho > 1$. Here we have $s-2$ eigenvalues, $\left\{ c_j^2 \middle| 3 \leq j \leq s \right\}$ of $gg^\dagger$ that are doubly degenerate and one fourfold degenerate eigenvalue, c_1^2, of $gg^\dagger$, which is associated with two doubly degenerate eigenvalues, $\{c_1, -c_1\}$ of g affiliated with CGSO's which can be linear combinations of $\{|\eta_1\rangle, |\eta_2\rangle\}$ and $\{|\eta_{-1}\rangle, |\eta_{-2}\rangle\}$ respectively [**?**]. In this case unitary transformations, $u(\tilde{\mathbf{z}}, \eta)$, belonging to $U(2s)$ can be expressed in factored form as

$$\begin{aligned} u(\tilde{\mathbf{z}}, \eta) &= c(\tilde{\mathbf{z}})\, k_\sigma(\eta_\sigma)\, k_\omega(\eta_\omega) \\ &= c_{+1}(\mathbf{z}_{\sigma+})\, u_{+1}(\mathbf{z}_{\omega+})\, u_0(\lambda_\omega)\, c_0(\lambda_\sigma)\, u_{-1}(\mathbf{z}_{\omega-})\, c_{-1}(\mathbf{z}_-)\, k_\sigma(\eta_\sigma)\, k_\omega(\eta_\omega), \end{aligned} \tag{45}$$

where $k_\sigma(\eta_\sigma) \in SU(2)^{s-2}$ and $c_{\sigma 0}(\lambda_\sigma) \in \exp N_0$ and are expanded in the basis eqs.$(37b)$ and $(37d)$ respectively with the operators referring to CGSO's $\{|\eta_j\rangle | j = \pm 1, \pm 2\}$ deleted. The operators $c_{+1}(\mathbf{z}_{\sigma+}) \in \exp N_{+1}$ and $c_{-1}(\mathbf{z}_{\sigma-}) \in \exp N_{-1}$ are expanded in the basis eqs.$(37a)$ and $(37c)$ with the operators referring to the pairs $\{(|\eta_j\rangle, |\eta_k\rangle) | j = \pm 1, k = \pm 1\}$ of CGSO's deleted.

The factors $u_{+1}(\mathbf{z}_{\omega+}) \in \exp U_{+11}, u_{-1}(\mathbf{z}_{\omega-}) \in \exp U_{-11}, u_{\omega 0}(\lambda_\omega) \in \exp U_{01}, k_\omega(\eta_\omega) \in USp(4)$, where U_{+1}, U_0, U_{-1} and $usp(4)$ are expanded in the following bases

$$U_{+11} = \mathbb{C}-\text{ linspan}\left\{a_k^\dagger a_j + \sigma(j)\,\sigma(k)\,a_{-j}^\dagger a_{-k};\quad j=1,-1;k=2,-2\right\} \tag{46}$$

$$\begin{aligned}U_{01} = \mathbb{C}-\text{ linspan}&\left\{\left(a_j^\dagger a_j + a_{-j}^\dagger a_{-j}\right), \left(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}\right), \left(a_j^\dagger a_{-j} + a_{-j}^\dagger a_j\right),\right.\\
&i\left(a_j^\dagger a_{-j} - a_{-j}^\dagger a_j\right), i\left(a_k^\dagger a_j - \sigma(j)\,\sigma(k)\,a_{-j}^\dagger a_{-k} - a_j^\dagger a_k + \sigma(j)\,\sigma(k)\,a_{-k}^\dagger a_{-j}\right),\\
&\left(a_k^\dagger a_j - \sigma(j)\,\sigma(k)\,a_{-j}^\dagger a_{-k} + a_j^\dagger a_k - \sigma(j)\,\sigma(k)\,a_{-k}^\dagger a_{-j}\right);\\
&\left. \qquad\qquad\qquad\qquad\qquad\qquad j=1,-1;k=2,-2\right\}\end{aligned} \tag{47}$$

$$U_{-11} = \mathbb{C}-\text{ linspan}\left\{a_k^\dagger a_j - \sigma(j)\,\sigma(k)\,a_{-j}^\dagger a_{-k};\quad j=1,-1;k=2,-2\right\} \tag{48}$$

$$\begin{aligned}usp(2\omega_1) &= usp(4)\\
&= \mathbb{R}-\text{ linspan}\left\{i\left(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}\right), i\left(a_j^\dagger a_{-j} + a_{-j}^\dagger a_j\right), \left(a_j^\dagger a_{-j} - a_{-j}^\dagger a_j\right),\right.\\
&\qquad\left(a_k^\dagger a_j - \sigma(j)\,\sigma(k)\,a_{-j}^\dagger a_{-k} - a_j^\dagger a_k + \sigma(j)\,\sigma(k)\,a_{-k}^\dagger a_{-j}\right),\\
&\qquad i\left(a_k^\dagger a_j - \sigma(j)\,\sigma(k)\,a_{-j}^\dagger a_{-k} + a_j^\dagger a_k - \sigma(j)\,\sigma(k)\,a_{-k}^\dagger a_{-j}\right);\\
&\qquad\qquad\qquad\qquad\qquad\left. j=1,-1;k=2,-2\right\},\end{aligned} \tag{49}$$

where $\sigma(j) = \text{sign}(j)$. The manifold of $U(2s)/\left[USp(4)\times SU(2)^{s-1}\right]$-coherent states is then given by

$$|\Phi(\mathbf{z})\rangle = c_{+1}(\mathbf{z}_{+\sigma})\,u_{+1}(\mathbf{z}_{+\omega})\,u_{01}(\lambda_{\omega_1})\,c_0(\lambda_\sigma)\left|g^N\right\rangle, \tag{50}$$

where $\mathbf{z} \equiv (\mathbf{z}_{+\sigma}, \mathbf{z}_{+\omega}, \lambda_\omega, \lambda_\sigma)$ and the reference AGP is generated by a geminal $|g\rangle$ in the class $\{s-2,2,0\}$.

If $\rho > 2$ then one deletes the appropriate operators in the bases $\{N_{+1}, N_0, N_{-1}, \mathcal{K}\}$ defined in eqs. $(37a)-(37d)$ and defines new bases $\{\{U_{+1j}, U_{0j}, U_{-1j}, usp(2\omega_j)\}|\,1\le j\le\rho\}$ according to eqs. $(46)-(49)$. The subalgebras $\{U_{-1j}, usp(2\omega_j)|\,1\le j\le\rho\}$ annihilate $\left|g^N\right\rangle$ i.e.

$$b\left|g^N\right\rangle = 0 \quad \forall b \in \bigoplus_{1\le j\le\rho} U_{-1j} \oplus \bigoplus_{1\le j\le\rho} usp(2\omega_j), \tag{51}$$

while the subalgebras $\{U_{+1j}, U_{0j}|\,1\le j\le\rho\}$ have the property

$$b\left|g^N\right\rangle \neq 0 \quad \forall b \in \bigoplus_{1\le j\le\rho} U_{+1j} \oplus \bigoplus_{1\le j\le\rho} U_{0j}. \tag{52}$$

The $U(2s)/\left[\prod_{1\leq j\leq\rho} USp(2\omega_j)\times SU(2)^{s-\omega}\right]$-coherent state manifold, $\mathcal{M}(s-\omega,\omega_1,\cdots,\omega_\rho,\xi)$, is then given by

$$|\Phi(\mathbf{z})\rangle = c_{+1}(\mathbf{z}_{\sigma+})\prod_{1\leq j\leq\rho}\{u_{+1j}(\mathbf{z}_{+j})\,u_{0j}(\lambda_j)\}\,c_0(\lambda_\sigma)\left|g^N\right\rangle. \tag{53}$$

The manifold of extreme AGP's, $\mathcal{M}(0,s,0)$, are $U(2s)/USp(2s)$-coherent state, which have been discussed in super conductivity [38] is given by

$$|\Phi(\mathbf{z})\rangle = u_{+11}(\mathbf{z}_{+1})\,u_{01}(\lambda_1)\left|g^N\right\rangle. \tag{54}$$

F. $SU(2\xi)$ factors

The changes from $SU(2)^s$ can this time be illustrated by an $\sigma=s-1,\rho=0,\xi=1$ example, ($|g\rangle$ belongs the class $\{s-1,0,1\}$), which also clearly leads to the general picture for $\xi>1$. Here we have $s-1$ eigenvalues, $\left\{c_j^2\middle|\,2\leq j\leq s\right\}$ of $gg^\dagger$ that are doubly degenerate and a doubly degenerate zero eigenvalue of $gg^\dagger$, which is associated with the CGSO's $\{|\eta_{-1}\rangle,|\eta_1\rangle\}$. In this case unitary transformations, $u(\tilde{\mathbf{z}},\eta)$, belonging to $U(2s)$ can be expressed in factored form as

$$u(\tilde{\mathbf{z}},\eta) = c(\tilde{\mathbf{z}})\,k_\sigma(\eta_\sigma)\,k_\xi(\eta_\xi) = c_{+1}(\mathbf{z}_{\sigma+})\,v_0(\lambda_\xi)\,c_0(\lambda_\sigma)\,c_{-1}(\mathbf{z}_-)\,k_\sigma(\eta_\sigma)\,k_\xi(\eta_\xi)\,, \tag{55}$$

where $k_\sigma(\eta_\sigma)\in SU(2)^{s-1}$ and $c_0(\lambda_\sigma)\in\exp N_0$ which are expanded in the basis eqs.(37b), (37d) with the operators referring to CGSO's $\{|\eta_j\rangle|\,j=\pm1\}$ deleted. The definitions of $c_{+1}(\mathbf{z}_{\sigma+})\in\exp N_{+1}$ and $c_{-1}(\mathbf{z}_{\sigma-})\in\exp N_{-1}$ are unchanged and are still expanded in the basis eqs.(37a) and (37c). The factors $v_0(\lambda_\xi)\in\exp V_0, k_\xi(\eta_\xi)\in SU(2)$, where $su(2)$ is expanded in the following bases

$$su(2) = \mathbb{R}-\text{linspan}\left\{i\left(a_1^\dagger a_{-1}+a_{-1}^\dagger a_1\right),i\left(a_1^\dagger a_1-a_{-1}^\dagger a_{-1}\right),\left(a_1^\dagger a_{-1}-a_{-1}^\dagger a_1\right)\right\} \tag{56}$$

and V_0 is given by

$$\{su(2)\oplus\mathcal{N}\}^{\mathbb{C}}, \tag{57}$$

where

$$\mathcal{N} = \mathbb{R}-\text{linspan}\left\{\left(a_1^\dagger a_1+a_{-1}^\dagger a_{-1}\right)\right\}. \tag{58}$$

The subalgebra $\{su(2\xi)\}$ annihilates $\left|g^N\right\rangle$ i.e.

$$b\left|g^N\right\rangle = 0 \quad \forall b\in su(2\xi)\,. \tag{59}$$

As the class of the reference AGP changes from $\{s-1,0,1\}$ to$\{s-p,0,p\}$ it is clear how the definitions of the factors eq.(55) evolve. It is easy to confirm that the class $\{M, 0, s-M\}$ cannot in fact exist, while the class $\{0, M, s-M\}$ does and such reference AGP's are IPS's and the $U(2s)/[USp(2M) \times SU(2s-2M)]$-coherent state manifold is in fact the Thouless $U(2s)/[U(2M) \times U(2s-2M)]$-coherent state manifold.

G. $U\left(\mathcal{H}^N\right)/U(1)$-Coherent States

It is illuminating and instructive to consider the CS formulation of the manifold, $\mathcal{M}_N$, of all N-particle states based on the unitary group $U\left(\mathcal{H}^N\right) \equiv U\left(\binom{r}{N}\right)$ of $\mathcal{H}^N$. In this case the configuration space formed by the coherent sates represents all of the states $|\Psi\rangle \in \mathcal{H}^N$ via a parameterization $\{|\Phi(\mathbf{z})\rangle\}$ and the *exact* quantum evolution equations can be expressed in terms of classical equations of motion, that are equivalent to the full TDSE, for paths belonging to $\mathcal{M}_N$. The isotropy subgroup is

$$\begin{aligned} K &= \left\{ g |\, g \in G, g\,|\Phi_1\rangle = e^{i\theta(g)}\,|\Phi_1\rangle \right\} \\ &= \left\{ \exp\left(i\lambda(g)\,|\Phi_1\rangle\langle\Phi_1|\right) \exp\left(i \sum_{i,j\neq 1} \lambda_{ij}(g)\,|\Phi_i\rangle\langle\Phi_j| \right) \right\}. \end{aligned} \tag{60}$$

The coset representatives are

$$U\left(\mathcal{H}^N\right)/K = \left\{ \exp\left(\sum_{i>1} \eta_i\,|\Phi_i\rangle\langle\Phi_1| \right) \exp\left(\eta_1\,|\Phi_1\rangle\langle\Phi_1|\right) \exp\left(-\sum_{i>1} \bar{\eta}_i\,|\Phi_i\rangle\langle\Phi_1| \right) \right\}, \tag{61}$$

where $\left\{ |\Phi_i\rangle | \quad 1 \le i \le \binom{r}{N} \right\}$ is a complete orthonormal basis for $\mathcal{H}^N$. The family of CS's associated with this coset is defined as

$$|\Phi(\eta)\rangle = \left\{ \exp\left(\sum_{i>1} \eta_i\,|\Phi_i\rangle\langle\Phi_1| \right) \exp\left(\eta_1\,|\Phi_i\rangle\langle\Phi_1|\right) \right\} |\Phi_1\rangle\,. \tag{62}$$

It is convenient to introduce a new equivalent coordinate system $\{\mathbf{z}\}$ by setting

$$\begin{aligned} &\eta_1 = \ln(z_1) \Leftrightarrow z_1 = e^{\eta_1} \\ &\eta_j = \frac{z_j}{z_1}; \quad 2 \le j \le \binom{r}{N} \Leftrightarrow z_j = \eta_j e^{\eta_1}, \end{aligned} \tag{63}$$

which leads to the following form for the coherent states

$$|\Phi(\mathbf{z})\rangle = \sum_{1 \le j \le \binom{r}{N}} z_j\,|\Phi_j\rangle\,. \tag{64}$$

Clearly eq. (64) is the familiar CI expansion of a general N-fermion state and the parameters $\{z_j\}$ the configurational interaction coefficients. The normalization of these CS's is given by

$$\langle \Phi(\mathbf{z}) | \Phi(\mathbf{z}) \rangle = \sum_{1 \le j \le \binom{r}{N}} |z_j|^2 = |z_1|^2 + \sum_{2 \le j \le \binom{r}{N}} |z_j|^2 \tag{65}$$

$$\langle \Phi(\eta) | \Phi(\eta) \rangle = e^{2\,\mathrm{Re}\,\eta_1} \left\{ 1 + \sum_{2 \le j \le \binom{r}{N}} |\eta_j|^2 \right\}. \tag{66}$$

Hence if one sets η_1 as the function

$$\eta_1 = \eta_1\left(\eta_2, \cdots, \eta_{\binom{r}{N}}\right) = \frac{1}{2} \ln \left(1 - \sum_{2 \le j \le \binom{r}{N}} e^{2\,\mathrm{Re}\,\eta_j} \right) \tag{67}$$

and z_1 as the function

$$z_1 = z_1\left(z_2, \cdots, z_{\binom{r}{N}}\right) = \left(1 - \sum_{2 \le j \le \binom{r}{N}} |z_j|^2 \right)^{\frac{1}{2}} \tag{68}$$

the family of CS's $\{|\Phi(\mathbf{z})\rangle\} \equiv \{|\Phi(\eta)\rangle\}$ are normalized if the reference state $|\Phi_1\rangle$ is normalized i.e. $\langle \Phi_1 | \Phi_1 \rangle = 1$. If one uses the CS parameters $\left\{\eta_2, \cdots, \eta_{\binom{r}{N}}\right\}$ or $\left\{z_2, \cdots, z_{\binom{r}{N}}\right\}$ the CS manifold is the unit sphere in $\mathcal{H}^N$, while if one uses $\{\eta\}$ or $\{\mathbf{z}\}$ then it is all of $\mathcal{H}^N$ and one explicitly considers the normalization in all expressions.

V. SUMMARY AND DISCUSSION

We have described how there are three possible chains of coset decompositions of the one particle unitary group $U(r)$ starting from maximal compact subgroups. Each of these chains produce families of cosets $\{U(r)/K\}$. A coset produces a manifold of coherent states from a reference state $|\Phi_{ref}\rangle$ if this state carries a one dimensional unitary representation of the subgroup K and the Hilbert space that $|\Phi_{ref}\rangle$ belongs to carries an irreducible representation of $U(r)$. As we are considering $N = 2M$ electron states this condition rules out cosets based on the maximal compact groups $SO(r)$ and $U(p) \times U(r-p)$ for $p \neq N$, leaving only $USp(2s)$ and $U(N) \times U(r-N)$. The subgroups $USp(2\omega_1) \times \cdots \times USp(2\omega_p) \times SU(2)^{\sigma} \times SU(2\xi)$ of $USp(2s)$ also satisfy this

CS condition, but there are no subgroups of $U(N) \times U(r-N)$ that do. The reference state for all of these cosets is an AGP state belonging to the class (see section III) $\{\sigma, \omega_1, \cdots, \omega_\rho, \xi\}$ that is determined by the canonical coefficients $\{c_j\}$ of the geminal $|g\rangle$, this index set also classifies the possible isotropy subgroups and hence the cosets. The Lie algebra of $U(r)$ has different direct sum decompositions determined by the different subgroups generating the cosets. The manifolds of AGP CS's $\mathcal{M}(\sigma, \omega_1, \cdots, \omega_\rho, \xi)$ have complex coordinates systems $\mathbf{z}$ wrt the reference state dependent bases adapted to these Lie algebra decompositions and these coordinates parameterize AGP states. Different manifolds contain AGP states that describe different physical properties e.g. extreme AGP's have properties associated with highly correlated systems, such as superconductors.

The manifolds $\mathcal{M}(\sigma, \omega_1, \cdots, \omega_\rho, \xi)$ equipped with the coordinate system $\mathbf{z}$ are particularly useful as they have a natural generalized phase space structure and can be used to construct, via the TDVP, approximations of the TDSE as a classical dynamical system. If one used the manifold constituted by whole Hilbert space $\mathcal{H}^N$ (not an AGP manifold) then the classical Hamilton equations plus an equation for a time dependent phase would be equivalent to the TDSE, and their solutions would provide solutions to the Schrödinger equation. The classical equations restricted to the AGP manifolds produce paths of AGP states that approximate the exact paths. One can consider the classical linear response of approximate stationary states found on these manifolds to external time dependent perturbations either within the curved phase or in the tangent space at these points [36]. This allows a variety of linear response approximations schemes to be developed that generalize and clarify propagator and Random Phase approximations.

APPENDIX A: APPENDIX: AGP STATES

The $N = 2M$ electron AGP states, $|g^N\rangle$ can be expanded as

$$\left|g^N\right\rangle = \sum_{1\le j_1<j_2<\cdots<j_N\le s} c_{j_1}\cdots c_{j_N}\left|\eta_{j_1}\eta_{j_1+s}\cdots\eta_{j_N}\eta_{j_N+s}\right\rangle, \tag{A1}$$

where $\{|\eta_j\rangle|\, 1 \le j \le r = 2s\}$ are Canonical General Spin Orbitals (CGSO) which form a complete orthonormal basis for $\mathcal{H}^1$, and $\{c_j|\, 1 \le j \le s\}$ are positive real numbers called the canonical coefficients of the two electron geminal $|g\rangle$ that has the canonical expansion

$$|g\rangle = \sum_{1\le j\le s} c_j \left|\eta_j\eta_{j+s}\right\rangle. \tag{A2}$$

The canonical coefficients are the square roots of the occupation numbers $\{n_j = n_{j+s}|\, 1 \le j \le s\}$ of the First Order Reduced Density Operator

(FORDO) $D^1(g)$ of the two electron state $|g\rangle$. The number, ν, of non zero $\{n_j|\, 1 \leq j \leq s\}$ is called the rank of the geminal $|g\rangle$, the dimension of the null space of $|g\rangle$ .is equal to $2(s-\nu)$. If all the occupation are equal (hence nonzero) the geminal is called extreme and the state $\left|g^N\right\rangle$ an extreme AGP state.

The CGSO's $\{|\eta_j\rangle|\, 1 \leq j \leq r\}$ diagonalize both the FORDO of the geminal state $|g\rangle$ and the AGP state $\left|g^N\right\rangle$ and are thus Natural General Spin Orbitals (NGSO's) of these operators, *however* NGSO's of these operators *are not* necessarily CGSO's! In fact if one multiplies $\{|\eta_j\rangle|\, 1 \leq j \leq r\}$ by arbitrary, but at least some non equal, phase factors a *different* geminal $|g\rangle$ is produced. It is interesting to observe that different geminals produced in such a manner have the same FORDO.

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